

Parth K. Thaker

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Education

Arizona State University

M.S. & Ph.D. in Electrical Engineering

Phoenix, US

Aug. 2017 – May 2024

Indian Institute of Technology, Madras

B.Tech. & M.Tech. in Electrical Engineering;

Chennai, IN

Aug. 2011 – Aug. 2016

Research Experience

Enhancing DINO based Video search

- Analysed and identified the shortcomings of DINO-based video search tool.
- Modified the training routine for the video search tool to enable improved fine-grained search capabilities
- Provided a quality-based assessment of the due to limited data and laid a road map for future directions for a more thorough evaluation and improvements to the already implemented enhancements.

Bandit-based multi-agent resource monitoring

- Proposed an algorithm that combines thresholding MAB with grid-based search with fixed search patterns to achieve successful search termination with bounds on the economic costs and completion times.
- Proposed two metrics, *priority labeling time* and *economic cost*, to study the performance of the proposed algorithm. We demonstrate the minimax-optimality of the proposed algorithm based on these metrics.
- Integrated coordination requirements and the physical limitations of switching actions directly into the algorithm. We demonstrate the efficacy of our approach using numerical simulations.

Bandits with graph structure

- Proposed a novel algorithm **GRUB**(**GR**aph **UCB** based Action Elimination) capable of capitalizing on structural graph information in Best-arm identification in stochastic bandits.
- Established rigorous theoretical complexity guarantees for GRUB showcasing the dependence of a graph-based speedup factor (scales as $\Omega(\text{clusters})$).
- Modular python implementation of GRUB is available at this Git repository with accompanying evidence of performance boost compared to baseline algorithm.

Nonconvex Landscape of Quadratic Feasibility

- Identified a subclass of Quadratic Programs, generally non-convex and NP-hard to solve, which can be tackled in a computationally efficient manner by first-order gradient descent methods.
- Theoretically established, under sufficiency conditions, that a non-convex loss function surrogate for the said QCQP satisfy *all local minima are in-fact global minima* and *all saddle points have strict negative curvature* property to guarantee success.
- Provided order-optimal sample complexity bounds in terms of the number of measurements for solving quadratic feasibility problems.

Differential programming using hyperspectral unmixing

- Incorporated a physics-based spectral variation model into a spectral unmixing pipeline to obtain superior performance.
- Part of a multi-departmental team to draw insights from optimization theory, physics and Deep learning methodology to propose a sequence of experiments to be performed for tackling the spectral unmixing problem.
- Provided conditions for initialization and theoretical convergence of alternate minimization approaches for spectral unmixing.

Sensor Fusion

- Developed module to determine the deviation of the real-time orientation of mounted devices as well as detecting aggressive driving patterns (Hard acceleration, Hard braking, heavy swirling, etc.) using inertial sensor data.
- Worked on end-to-end implementation of the inertial modules including implementation of data acquisition algorithms from onboard inertial sensors, analysis of acquired data and notifying alerts on mobile and web applications.

Factored gradient descent

- In most real-world applications, projection is generally a computationally intensive operation. Proposed a cost-efficient variant of projected gradient descent by splitting the gradient step and projection step into two timescale update algorithms. Performed experimental evaluation of trade-off for the proposed method.
- Extended factored gradient descent methods to tackle the problem of low rank estimation in fat and tall matrices using alternate minimization routines.

Work Experience

Inuitive Surgical | Intern

Sunnyvale | May 2023 - Aug 2023

- Enhanced the quality of suggestions of DINO-based Video Search tool by adding fine-grained learning capabilities during self-supervised training routines.

Mitsubishi Electric Research Laboratories | Intern

Boston | May 2022 - Aug 2022

- Developed bandit algorithms for the task of resource monitoring with experimental validation.

Netradyne | Systems Engineer

Bangalore | Aug 2016 - May 2017

- Worked on data pipelineing (acquisition, cleaning and analytics) for reckless driving alerts using inertial sensors.

Securifi Systems | Intern

Hyderabad | May 2014 - Aug 2014

- Worked on server infrastructure automation and management using Puppet master slave architecture.
- Evaluated test metrics for comparing performance of Cassandra, MongoDB and MySQL on specific data formats.

Cisco Systems Pvt. Ltd | Intern

Bangalore | May 2013 - Aug 2013

- Worked on python scripts for testing and overloading new proposed routers models using fake IP creation and bit-level manipulation of network signals.

Dissertation based Publications

Published

- **Parth Thaker**, Vineet Gattani, Vignesh Tirukkonda, Pouria Saidi, Gautam Dasarathy. “Non-stationary bandits with periodic behavior: Harnessing Ramanujan periodicity transforms to conquer time-varying challenges.” IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP 2024), Under review
- **Parth Thaker**, Stefano Di Cairano, Abraham P. Vinod. “Bandit-based multi-agent search under noisy observations.” 22nd IFAC World Congress, 2023.
- **Parth Thaker**, Mohit Malu, Nikhil Rao, Gautam Dasarathy. “Maximizing and Satisficing in Multi-armed Bandits with Graph Information”, Neural Information Processing Systems (NeurIPS), 2022.
- **Parth Thaker**, Gautam Dasarathy, and Angelia Nedić. “On the sample complexity and optimization landscape for quadratic feasibility problems.” In IEEE International Symposium on Information Theory (ISIT), 2020.

Working Manuscript

- **Parth Thaker**, Gautam Dasarathy, and Angelia Nedić. “The Nonconvex Optimization Landscape of Quadratic Feasibility.” arXiv preprint 2002.01066.
- **Parth Thaker**, Abraham Vinod, Gautam Dasarathy, Marcus Greiff, Stefano Di Cairano. “Multi-Armed Bandit-Based Adaptive Resource Monitoring with Constrained Mobile Robots.” In preparation.

Other Publications

Published

- John Janiczek, **Parth Thaker**, Gautam Dasarathy, Christopher Edwards, Philip Christensen, and Suren Jayasuriya. “Differentiable Programming for Hyperspectral Unmixing using a Physics-based Dispersion Model.” In 16th European Conference on Computer Vision (ECCV), 2020. Springer International Publishing.
- **Parth Thaker**, Aditya Gopalan, and Rahul Vaze. “When to arrive in a congested system: Achieving equilibrium via a learning algorithm.” In the 15th International Symposium on Modeling and Optimization in Mobile, Ad Hoc, and Wireless Networks (WiOpt). IEEE, 2017.

Miscellaneous

Awards

- Awarded Engineering Graduate Fellowship for the 2019-2020 academic year by the Ira A. Fulton Schools of Engineering.

Coursework

- Real Analysis, Functional Analysis, Applied Probability, Large-scale Optimization, Graph Theory.
- Statistical Machine learning, Process optimization, Computation methods in EE, Multivariate Data Analysis.

Skills & Tools:

- Python, Bash scripting, MySQL, Cassandra, OpenCV.